DOM Manipulation in JavaScript Assignment

PW EARNERS : Coding and Web Development

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Topics Covered

- DOM Tree Structure
- window vs document
- getElementById()
- querySelector()
- querySelectorAll()
- innerHTML
- textContent
- setAttribute()
- classList
- style
- dataset
- createElement()
- appendChild()
- append()
- prepend()
- insertBefore()
- removeChild()
- remove()
- cloneNode()

Section 1 - Selecting and Modifying Elements

1. Select an Element by ID

Create a heading with an id of title and use getElementById() to select it and change its text.

Example:
HTML: <h1 id="title">Welcome</h1>
Output: Hello JavaScript

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2. Select an Element Using querySelector()

Create a paragraph with a class description and use querySelector() to select it and change its text content.

Example:
HTML: Old Text
Output: New Description

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3. Select Multiple Elements Using querySelectorAll()

Create three <li> elements with the class item. Use querySelectorAll() to select all of them and change their text color using the style property.

Example:

HTML: <li class="item">HTML

<li class="item">CSS

<li class="item">JavaScript

Expected Result: All three list items should have the changed text color.

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4. Change Content Using textContent

Create a paragraph containing some text and use textContent to replace its content with a new message.

Example:

Before: Old Message

After: Welcome to JavaScript!

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5. Add HTML Using innerHTML

Create a with an id of container and use innerHTML to add a heading and a paragraph inside it.

Example:

Expected HTML inside the container:

<h2>My Website</h2>

Welcome to my website!

Section 2 - Attributes, Classes and Styles

6. Change an Attribute Using setAttribute()

Create an image element and use setAttribute() to change its src and alt attributes.

Example:

HTML: <img id="profileImage" src="old.jpg" alt="Old Image">

Expected Result: The image should have the new src and alt values.

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7. Add and Remove Classes Using classList

Create a button and use classList.add() to add a class to it. Then use classList.remove() to remove the class.

Example:

HTML: <button id="btn">Click Me</button>

Expected Result: The button should have the class added and then removed using JavaScript.

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8. Modify Element Style

Create a heading and use the style property to change its color, fontSize, and backgroundColor.

Example:

HTML: <h1 id="heading">JavaScript</h1>

Expected Result: The heading should display with the new styles applied through JavaScript.

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9. Read Data Using dataset

Create a button with a custom data-id attribute and use the dataset property to read its value.

Example:

HTML: <button id="productBtn" data-id="101">View Product</button>

Output: 101

Section 3 - Creating and Adding Elements

10. Create an Element Using createElement()

Create a new <p> element using createElement(), add some text to it using textContent, and display it on the webpage.

Example:

Expected Output: This paragraph was created using JavaScript.

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11. Add an Element Using appendChild()

Create a <ul> in HTML. Use JavaScript to create a new <li> element and add it to the list using appendChild().

Example:

HTML: <ul id="skills">

Expected Output: HTML, CSS, JavaScript

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12. Add Elements Using append() and prepend()

Create a list and use append() to add an item at the end and prepend() to add an item at the beginning.

Example:

Before: CSS, JavaScript

After: HTML, CSS, JavaScript, React

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13. Insert an Element Using insertBefore()

Create a list containing three items and use insertBefore() to insert a new item before the second item.

Example:

Before: HTML, JavaScript, React

After: HTML, CSS, JavaScript, React

Section 4 - Removing and Cloning Elements

14. Remove an Element

Create a list containing three items and remove one item using either removeChild() or remove().

Example:

Before: HTML, CSS, JavaScript

After: HTML, JavaScript

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15. Clone an Element Using cloneNode()

Create a button and use cloneNode() to create a copy of the button. Add the cloned button to the webpage.

Example:

HTML: <button id="btn">Click Me</button>

Expected Result: [Click Me] [Click Me]